

Jake Hoover

EDUCATION

University of Alberta

Edmonton, AB, Canada

Doctorate in Philosophy (Thes) - Mathematical and Statistical Sciences (GPA 4.0)

September 2025 - Current

Select Courses: Stochastic Processes, Functional Data Analysis, Probabilistic Graphical Models.

University of Alberta

Edmonton, AB, Canada

Masters of Science (Thes) - Mathematical and Statistical Sciences (GPA 3.7)

September 2022 - August 2025

Select Courses: High Dimensional Probability, Regression Analysis, Statistics for Learning, Probability and Measure, Discrete Data Analysis, Statistical Computing, Methods of Statistical Inference, Hidden Markov Models for Time Series.

University of Alberta

Edmonton, AB, Canada

Bachelor of Science with a Specialization in Mathematics (GPA 3.3)

September 2015 - April 2021

Select Courses: Risk Theory, Theory of Complex Variables, Mathematical Analysis I & II, Time Series Analysis, Graph Theory, Probability, Probability and Stochastic Processes, Sampling Techniques, Differential Equations, Partial Differential Equations, Applied Regression Analysis, Mathematical Statistics, Design and Analysis of Experiments Linear Algebra I, II, & III, Coding Theory, Rings and Modules.

ACADEMIC EXPERIENCE

Teaching Assistant

University of Alberta (January 2023 - Current)

- Statistics Lab Lecturer for Introduction to Applied Statistics II (STAT 252, Spring 2025, Winter 2025), exploring fundamental statistical concepts such as hypothesis testing, ANOVA, and linear regression, using statistical software SPSS.
- Statistics Lab Lecturer for Introductory Statistics for Engineers (STAT 235, Summer 2025, Fall 2025, Winter 2026), covering a broad scope of statistical concepts from basic data analysis, probability, statistical distributions, hypothesis testing, and linear regression using excel.
- Curated the development of instructional videos for Introduction to Applied Statistics (STAT 151, Spring 2025, Summer 2025). Explored rudimentary data analysis and statistical concepts using SPSS.
- Select Student Feedback:
 - “The lab videos were useful and good quality.” - SPOT Student Survey Response STAT 252 (Summer 2025).
 - “He was very good at explaining the topics of the course and how to get what you wanted. He was very good at spending 1 on 1 time and answering any questions that I have had. He is the best TA I have had yet.” - SPOT Student Survey Response STAT 235 (Fall 2025).
 - “He broke down topics and allowed us to learn them in a way that was different and easier than in class. He did a very good job at being approachable and answering any questions that I had.” - SPOT Student Survey Response STAT 235 (Fall 2025).

Training and Consulting Centre

University of Alberta (January 2023 - Current)

- Meeting weekly with various clients to provide a range of services and assistance at any stage of the research process, including experiment and survey design, data collection, statistical modeling and data analysis, and result interpretation.
- Exposed to a wide variety of statistical problems that real world people are having, which emphasises what statistics is being used for outside of a research environment.

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Research Assistant

University of Alberta (January 2023 - Current)

- Conducting literature reviews and research summaries on a bi-weekly basis with a team of supervisors, sleep pathology experts, and fellow MSc students.
- Creating and running code for fitting Hidden Semi Markov Models on real world data with applications to polysomnography data sets.
- Engaging in interdisciplinary research groups in the Faculty of Dentistry, including with biomedical imaging teams where we use Convolutional Neural Networks to process images, and with professional dental health experts to leverage Natural Language Processing models to aid with medical charting.

Statistics and Mathematics Tutor

(2022 - Current)

- Working with university and high school students to help develop a deeper intuitive and technical understanding of statistical and mathematical sciences.

CONFERENCE PRESENTATIONS

Classification Society Annual Meeting 2026

Dalhousie University, Halifax, NS, Canada. June 17-18, 2026.

Poster Presentation: "General Dwell Time Hidden Semi-Markov Models: Is it Worth the Trouble?"

PROFESSIONAL EXPERIENCE

IT Infrastructure Intern

Cenovus (May 2023 - December 2023)

- Developed database solutions to facilitate statistical software development in the oil and gas industry.
- Assisted in the development of data cleaning, model fitting, and GUI interface creation for various statistical apps used in the company.

Materials Handler ATCO Warehousing

ATCO Gas (May 2020 - September 2022)

- Hands on experience in supply chain management of gas infrastructure hardware.
- Emphasis on project management and material acquisition and distribution in the field.

Projects and Construction Electrical Apprentice

ATCO Electric (May 2019 - August 2019)

- Stationed at Nisku fabrication warehouse, assembled electrical units to be deployed to substations across Alberta.
- Field deployed with ATCO's substation commissioning crews across the Northern Alberta region, worked with leading substation engineers and electricians to build substation infrastructure.

Operations and Maintenance Fixed Assets Intern

ATCO Electric (May 2016 - December 2017)

- Involved in the asset management and asset retirement programs.
- Implemented new asset retirement workflow to aid with the substantial backlog of formal processes.

SELECT SKILLS:

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- **Coding Languages:** R, Python, SPSS, Excel, RCmdr, Ansible.
- **Data Bases:** Mongo DB, Oracle, MySQL.
- **Cloud Architecture:** AZURE, AWS.

HONORS AND ACCOLADES:

- **Winner of “Best Phd Poster Presentation”** - Meeting of Alberta Statisticians 2026
- **Golden Key International Honour Society** (2024,2025,2026) - Recognized as a MSc student with an exceptional GPA
- **Full Time Research Assistantship** (University of Alberta 2024, 2025)
- **Full Time Teaching Assistantship** (University of Alberta 2024, 2025)
- **Student of the Month** (March 2026) - <https://www.instagram.com/p/DWcALrID2AP/>

ADDITIONAL ACTIVITIES

- Member of Statistical Society of Canada
- Member of Royal Statistical Society

VOLUNTEERING

Florence Nightingale Days - University of Alberta (*March 2025*)

- Introduced statistical concepts to high school students, with an emphasis on equity in STEM and increasing participation from underrepresented groups.
- Designed and delivered an interactive online notebook covering randomness, computer-generated random numbers, and how to leverage random members in applications.
- Demonstrated applied concepts such as numerical integration via Monte Carlo simulation to build intuition through hands-on exploration.
- [Workbook Link](#) - Take a look and learn a little about random numbers!

Community Dinners - Robertson-Wesley United Church (*2025 - Current*)

- Regularly volunteer at Community Dinner, assisting with meal preparation and serving food to community members in need. It is a rewarding experience that allows me to contribute in a practical way while helping create a warm, supportive environment.